

AI SYSTEMS ENGINEERING

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Objectives, Content, Material

OBJECTIVES

The course introduces the methodologies and tools for the development and evolution of AI systems, with reference to the entire life cycle: design, development, verification and validation, evolution/maintenance, operation. The course illustrates the peculiarities of AI systems compared to traditional systems, how these peculiarities impact on the various phases of the system life cycle, and which processes, models, languages, techniques, technologies and tools are required for high quality AI systems engineering.

CONTENT

Part I. Introduction.

Basic concepts. Definitions. AI Systems Engineering vs Data Science. AI System Functional Architecture and Architecture Implementation Framework. Life cycle of AI systems, development process.

Part II. Analysis and Design.

Agile practices, DevOps. Review of ML fundamentals. Building the ML workflow, Machine Learning Operations (MLOps). Stakeholder's business needs, requirements analysis. Design and Development: Design of a training system, data/feature engineering, model engineering. Training and re-training strategies, monitoring and drift detection. Automated training. Model persistence. Pipelining.

From development to deployment: the role of architectural patterns. Batch vs stream processing. Microservice architectures for ML systems. Deployment and operations. Deployment strategies. Blue-green, canary deployment. Support technologies: containers, Docker, Kubernetes. Example with MS Azure. Alternative technical solutions and tools.

Engineering Large Language Models (LLM) systems. LLM fundamentals. The LLM operations (LLMOps) cycle. Architectural and technical alternatives.

AI as a component: system-level design. System requirements modeling. Model-based System and Software Engineering. Simulation, Co-simulation. Modeling languages (SysML, UML). The example of Autonomous Driving Systems.

Part III. Testing and Quality Assurance

ML testing, verification and validation. Testing workflow, testing components. Quality attributes for AI systems: correctness, robustness, privacy, efficiency, fairness, interpretability. The ALTAI and FASTEPS classification. Responsible AI.

Testing of ML and Deep Neural Networks (DNN). Black box testing. White-box testing. The oracle problem. Coverage-guided DNN testing, operational DNN testing. Adversarial testing, Fuzz Testing. Mutation and Metamorphic testing. Adequacy criteria for DNN testing. Model in the loop: continuous testing of DNN. DNN Debugging and retraining.

Safety testing: examples of simulation-based safety testing for ADS. Security and privacy of AI systems: attacks and defences. Fairness testing techniques. Testing and evaluation of LLM. Open challenges. Standards, the EU AI Act.

Readings/Bibliography

Adopted Textbook

- 1) Machine Learning Engineering with Python – Second Edition: Manage the production life cycle of machine learning models using MLOps with practical examples, 2023, Andrew P. McMahon, Packt Publishing; ISBN: 978-1837631964
- 2) AI Engineering: Building Applications with Foundation Models. Chip Huyen, O'Reilly Media; ISBN: 978-1098166304

Further Suggested Textbook

- 1) Artificial Intelligence: A Systems Approach from Architecture Principles to Deployment, 2024, David R. Martinez, Bruke M. Kifle. The MIT Press; ISBN: 978-0262048989

Slides from lessons and exercises

Available on the Microsoft Team platform, and on the University teaching website:

<http://www.docenti.unina.it/roberto.pietrantuono>